Sam Doe

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Riverton, ST | Willing to relocate

Education

B.S. Electrical Engineering, GPA 3.45/4.00 Expected Spring 2029
Example State University
Relevant coursework: Electrical Circuits, Digital Logic, Foundational Programming Concepts and Design
Activities: IEEE student branch

Projects

Weather Station with Wireless Logging Spring 2026
ESP32, C, BME280, Python

- Logged BME280 data every minute for 30 days over Wi-Fi and lost under 1% of samples
- Calibrated temperature in C against a reference thermometer at 20 points, to within 0.5 °C
- Wrote a Python plotting script that flagged gaps over 5 minutes and found 3 Wi-Fi outages

4-Bit ALU on a Breadboard Fall 2026
74HC logic, Logisim

- Built a 4-bit ALU out of 14 74HC-series chips that can add, subtract, AND, OR and XOR, and checked it against a Logisim model for all 256 input pairs of each operation with DIP switches and LEDs
- Found 2 wiring faults in the carry chain by probing each stage with a logic analyzer

Experience

Electrical Lead, Example State Robotics Club, Riverton, ST Aug 2026–Present

- Designed and soldered a 2-layer power distribution board in KiCad with 6 fused outputs to replace the old hand-wired harness, so wiring before a match takes 45 minutes instead of 3 hours
- Traced controller resets on hard turns to a 1.2 V dip on the 5 V rail with an oscilloscope and gave the controller its own 5 V regulator, with no resets in the 12 matches since

Barista, Example Coffee Co., Riverton, ST May 2025–Present

Skills

Software: C, Python, KiCad, Logisim
Hardware/Lab: Oscilloscope, Logic analyzer, Soldering, ESP32